

Challenge 9 – Smart Industrial Pollution Monitoring for Golden Corridor (Vapi–Ankleswar)

Domain: Environmental Sustainability

The Challenge

The industrial belt along Vapi, Ankleshwar, and Vatva faces significant air and water pollution from chemical and manufacturing units, with limited real-time monitoring and slow regulatory response to violations.

Objective

Build Agentic AI solutions that continuously monitor industrial emissions and effluents, detect violations, and trigger timely regulatory and community alerts.

Suggested AI Agents

- Emission & Effluent Monitoring Agent
- Violation Detection & Compliance Agent
- Regulatory Alert & Escalation Agent
- Public Health Risk Assessment Agent
- Industrial Pollution Dashboard Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud